Task-5

Exploratory Data Analysis (EDA) Report on Titanic Dataset

1. Introduction

This report presents an Exploratory Data Analysis (EDA) of the Titanic dataset. The objective of the analysis was to understand the dataset, identify missing values, explore the distribution of variables, detect relationships among features, and extract meaningful insights using statistical summaries and visualization

2. Dataset Information

- **Dataset:** Titanic Dataset
- **Rows:** 891
- **Columns:** 12 (before cleaning)
- **Tools Used:**
  - Python
  - Jupyter Notebook
  - Pandas
  - Matplotlib
  - Seaborn

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3. Data Cleaning Summary

The following data cleaning steps were performed before the analysis:

- Loaded the dataset using Pandas.
- Checked the dataset structure using df.info().
- Generated statistical summaries using df.describe().
- Identified missing values using df.isnull().sum().
- Filled missing values in the **Age** column using the **median**.
- Filled missing values in the **Embarked** column using the **mode**.
- Removed the **Cabin** column because it contained a large number of missing values.
- Checked for duplicate records and confirmed the dataset was ready for analysis.

4. Exploratory Data Analysis

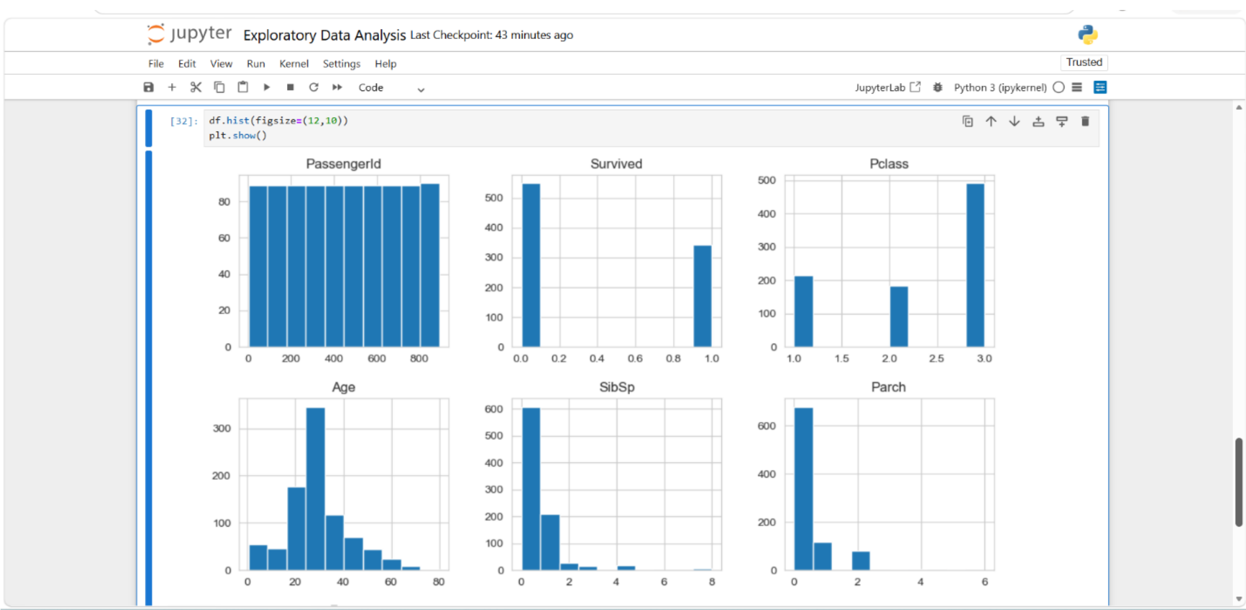
Several visualizations were created to understand the data.

Histogram

The histogram was used to analyze the distribution of numerical variables.

Observation:

- Most passengers were between **20 and 40 years** of age.
- The Fare distribution is positively skewed.

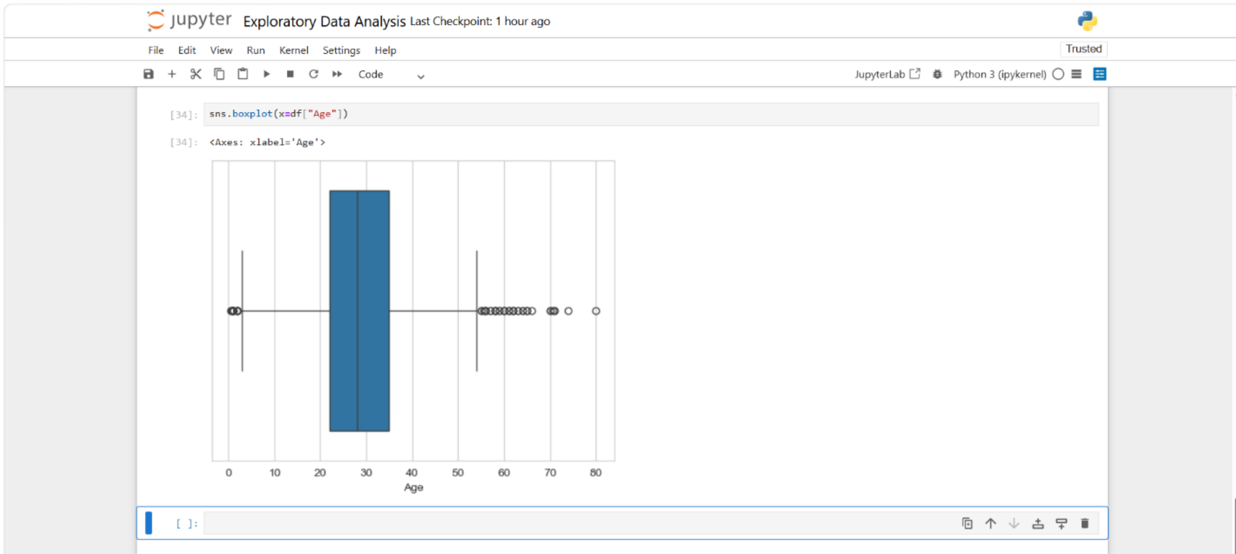


Boxplots

Boxplots were used to identify outliers in numerical features.

Observation:

- Age contains a few outliers.
- Fare contains several extreme outliers, indicating high variability.

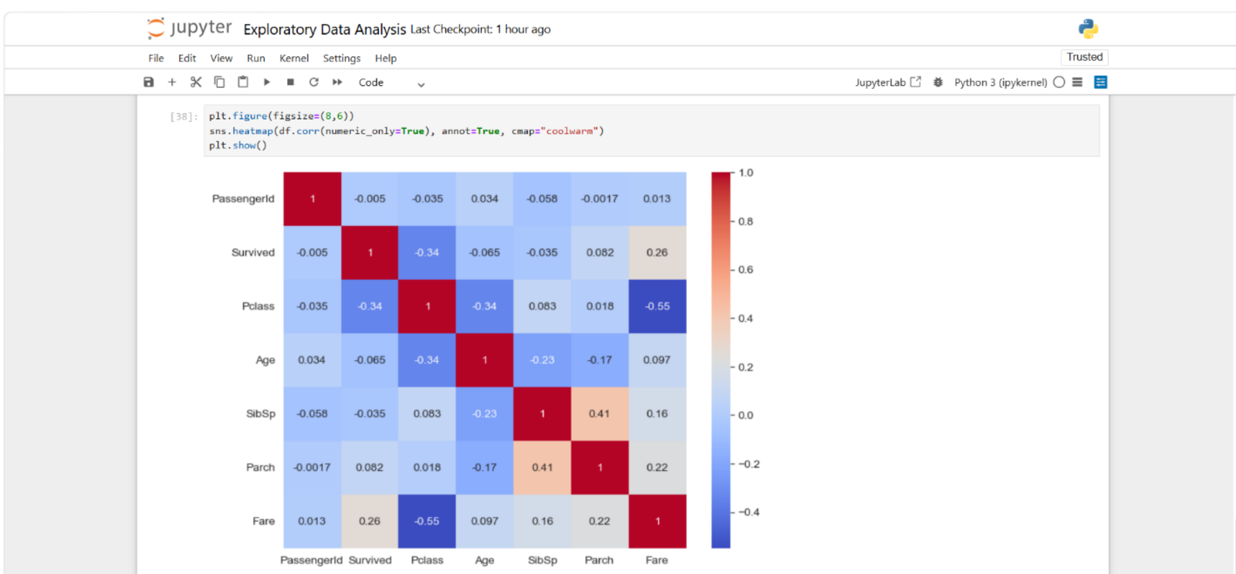


Correlation Heatmap

The heatmap was used to examine relationships among numerical variables.

Observation:

- Fare has a weak positive correlation with Survival.
- Passenger Class has a negative correlation with Survival.
- No strong multicollinearity was observed.

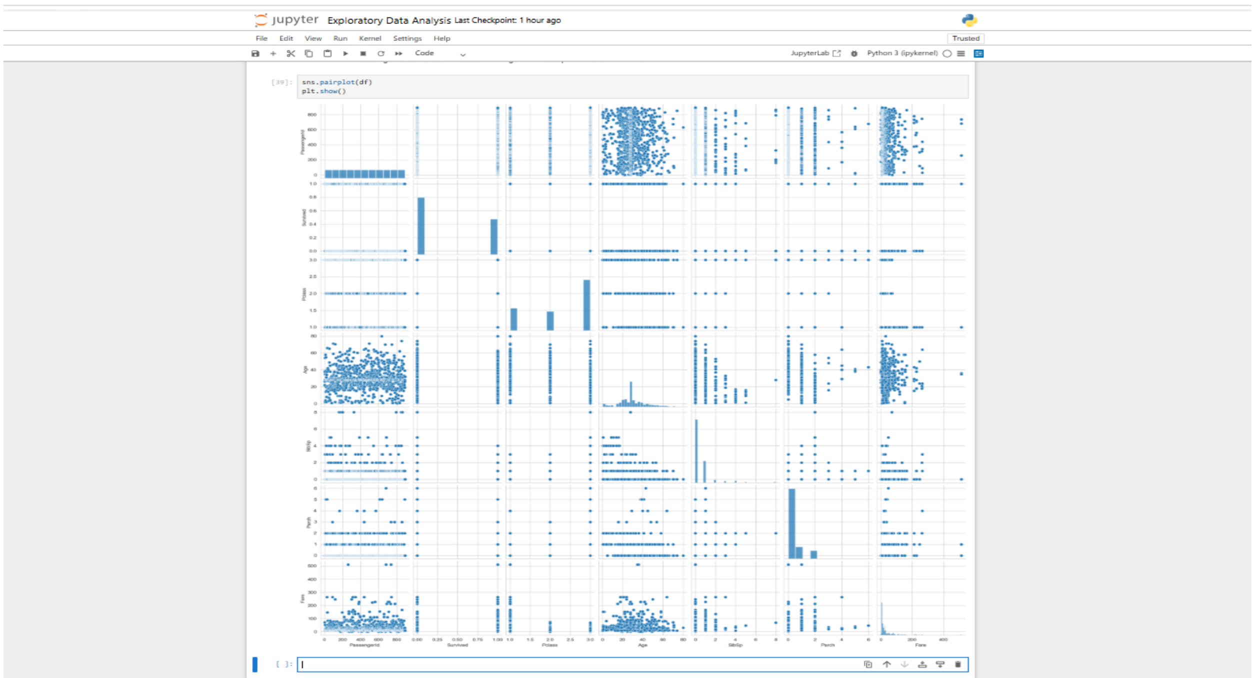


Pairplot

The pairplot helped visualize relationships between numerical variables.

Observation:

- Most numerical variables do not show strong linear relationships.
- The plot helps identify trends and potential outliers.

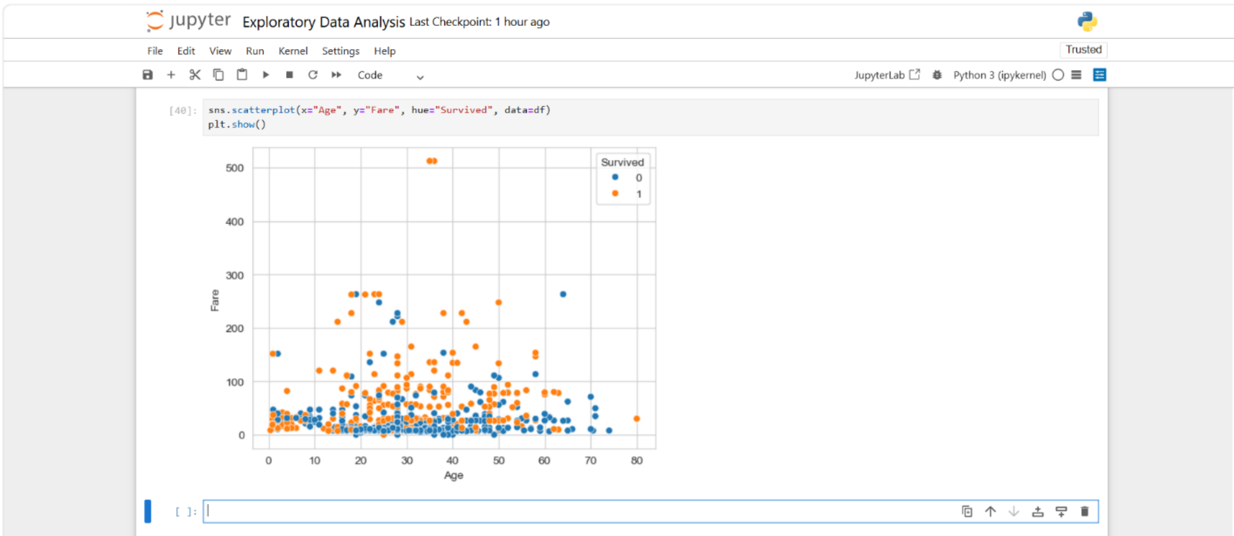


Scatter Plot

The scatter plot analyzed the relationship between Age and Fare.

Observation:

- No strong linear relationship exists between Age and Fare.
- Higher fare values are concentrated among fewer passengers.

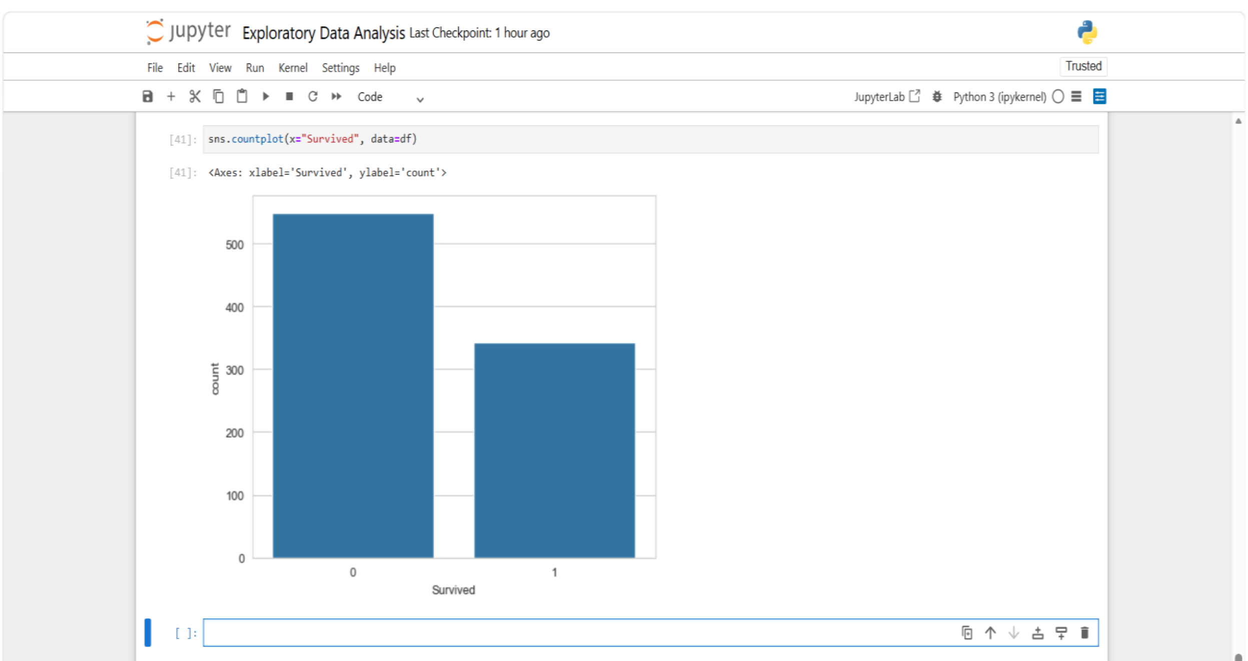


Survival Analysis

Countplots were used to compare survival based on Gender and Passenger Class.

Observation:

- Female passengers had a significantly higher survival rate than male passengers.
- First-class passengers had a better survival rate than second- and third-class passengers.



5. Key Findings

- Most passengers belonged to the **Third Class**.
- The majority of passengers were **20–40 years old**.
- The **Fare** variable is positively skewed and contains several outliers.
- Female passengers had a higher survival rate than male passengers.
- First-class passengers had a higher chance of survival.
- Passenger Class negatively correlates with survival.
- Fare shows a weak positive relationship with survival.

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6. Conclusion

The Exploratory Data Analysis successfully identified important patterns and relationships within the Titanic dataset. Data cleaning improved the dataset quality by handling missing values appropriately. Visualizations revealed that passenger class, gender, age, and fare influenced survival outcomes. Overall, EDA provided valuable insights into the dataset and created a strong foundation for further predictive modeling and machine learning applications.